

Deque Class Documentation

Source File: 'Deque.h'

Namespace: dsp

Class Header: template <class T> class Deque

Overview

The *Deque* class is a double-ended queue.

Constructors

- `Deque()` (default constructor)
 - **Purpose:** Constructs an empty deque.
- `Deque(const Deque<T>& obj)` (copy constructor)
 - **Purpose:** Makes a deep copy of *obj*.
 - **Parameter(s):**
 - *obj*: Constant *Deque* reference object.

Assignment Operators

- `operator=(const Deque<T>& rhs)`
 - **Purpose:** Makes a deep copy of *rhs*.
 - **Parameter(s):**
 - *rhs*: Constant *Deque* reference object.
 - **Return:** `*this`.

Destructor

- `~Deque()`
 - **Purpose:** Deallocates the deque.

Methods

- `insertFirst(const T& obj)`
 - **Purpose:** Inserts *obj* to the beginning of the deque.
 - **Parameter(s):**
 - *key*: Constant generic reference object.
- `insertLast(const T& obj)`
 - **Purpose:** Inserts *obj* to the end of the deque.
 - **Parameter(s):**
 - *key*: Constant generic reference object.
- `removeFirst()`
 - **Purpose:** Removes the front item from the deque.
- `removeLast()`
 - **Purpose:** Removes the back item from the deque.
- `first() const`
 - **Purpose:** Retrieves the front item of the deque.
 - **Exception(s):**
 - *out_of_range*: Occurs if the deque is empty.
 - **Return:** Constant generic reference object.
- `last() const`
 - **Purpose:** Retrieves the back item of the deque.
 - **Exception(s):**
 - *out_of_range*: Occurs if the deque is empty.
 - **Return:** Constant generic reference object.
- `empty() const`
 - **Purpose:** States if the deque is empty.
 - **Return:** A Boolean.